

Variability, its limits, and the performance-competence debate: Implications of linguistic variability for a theory of grammar

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Abstract

The article provides an overview of variability in natural language by looking at the limits of intra-individual, inter-individual, and cross-linguistic variability at all levels of grammar. We review evidence for the hypothesis that variability forms an integral part of natural language and often provides valuable insights into speakers' linguistic competence. We will discuss in turn (i.) different subtypes of variability; (ii.) the difference between systematic linguistic variability, as driven by grammar or processing-related factors, and random noise including performance errors; (iii.) the phenomenon of *hidden variability*, where linguistic expressions can differ in their underlying structure in spite of parallel surface strings; (iv.) the extent to which variability is accounted for in existing grammatical theories; (v.) variability in experimental quantitative data as a window into the underlying grammatical systems and processing mechanisms; (vi.) variability in language as an important trigger for diachronic change and language acquisition.

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1. INTRODUCTION

Natural language exhibits many instances of linguistic variability. Consider the English dative-alternation in example 1, as prominently discussed in Bresnan et al. (2007):

- (1) a. Joan gave [_{NP} the children] [_{NP} toys].
b. Joan gave [_{NP} toys] [_{PP} to the children].

Different linguistic schools have assigned different degrees of importance to linguistic variability and its limits in grammatical theory building. For instance, Chomsky (1965, p. 3) advocates to abstract away from random surface noise, such as performance errors, and inter- and intra-individual variability by focusing on the invariable primitives in the linguistic competence of speakers: “Linguistic theory is concerned primarily with an ideal speaker-listener, in a completely homogeneous speech-community, who knows its language perfectly and is unaffected by such grammatically irrelevant conditions as memory limitations, distractions, shifts of attention and interest, and errors (random or characteristic) in applying his knowledge of the language in actual performance”. At the other end of the scale, there are probabilistic approaches, such as Bresnan et al. (2007), which treat variability as an integral feature of language, and which use the observable variability in language to make a plea for a multifactorial probabilistic model of language, which postulates a greater space of grammatical possibilities than assumed in deterministic systems.

This review article provides an overview of linguistic variability and its limits, and how the systematic study of variability (or its absence) in language can contribute to the theory of mental grammar, i.e., the set of mental rules that constitute a speaker’s linguistic competence. By *linguistic variability* we understand the range of different possible linguistic behaviors in comprehension and production that are available to individual language users, between different language users, or between languages; see section 2.1. Another important distinction is that between performance- or processing-induced variability, on the one hand, and grammar-internal variability as part of speakers’ linguistic competence, on the other. Grammar-internal variability is evidenced by the possibility of more than one linguistic

behavior within and between speakers and between contexts, cf. example 1. By contrast, we will not consider cases of variability between different contexts. A prominent instance is the variable word order in discourse-configurational languages, which is predictably determined by information-structural factors (see, e.g., Kiss 1995).

We will review evidence for the hypothesis that variability is an integral part of natural language, and that many instances of variability in language open a window into the workings of language in the human mind. We will focus on different types of variability in section 2.1, the difference between systematic linguistic variability and random noise including *performance errors* in section 2.2, and *hidden variability*, relating to the fact that linguistic expressions can differ in their underlying structure in spite of parallel surface structures in section 2.3. We will then consider the extent to which the observable linguistic variability is integrated and accounted for in grammatical theories in section 3. Finally, we will stress the importance of intra- and inter-speaker variability in quantitative experimental data as a window into the underlying grammatical systems and processing mechanisms in section 4.1, and we will look at inter-speaker variability in language as a trigger of diachronic change and language acquisition in section 4.2.

The resulting picture assigns a considerable degree of flexibility to the grammars of natural languages. A set of grammatical core cases, which is shared across individuals or languages, is surrounded by a halo of optional possibilities, which are only available at times, or to some speakers, but which still differ from clearly ungrammatical cases. Grammatical theories must thus be equipped with devices to model this grammatically meaningful variability. In particular, grammatical models must be able to distinguish between those instances of variability that fall under Saussure's (1916) label of *langue* (competence), and those that follow from systematic (processing-driven) or unsystematic factors (*parole*/performance).

The paper adopts a similar perspective as Honeybone (2011) on variants of English and Yu & Zellou (2019) on variability in phonology, but it has a broader scope (i.) in looking at all levels of linguistic representations, (ii.) in considering (meaningful) variability at the group level, and (iii.) in focusing more on the implications for the modeling of mental grammar. Like Yu & Zellou (2019), we will set aside instances of variability caused by sociolinguistic and/or geographic components, which have been the subject of other ARL-papers, see, among others, Wieling & Nerbonne (2015); Thomas (2016); Gordon (2019); D'Alessandro (2021). Also, given space constraints, we do not aim to provide a comprehensive overview of all available work on the topic. Instead, we will focus on exemplary work that we consider to be representative of the issues discussed.

2. TYPES OF VARIABILITY IN LANGUAGE

This section introduces different subtypes of variability since grammatical theories (see section 3) differ as to which type the respective models can capture, and for which subtypes explanations must be sought in other (non-linguistic) components of the cognitive system.

2.1. Intra-Individual, Inter-Individual, and Cross-Linguistic Variability

We start by differentiating between levels at which variability can be observed: Linguistic behavior in comprehension and production can vary across or within languages, and in the latter case within speakers or across speaker (sub)groups.

Cross-linguistic variability is instantiated by linguistic phenomena involving different

structural configurations across languages. For instance, languages can differ in how they form constituent questions syntactically. *Wh-ex-situ* languages like English have the question word (here: direct object) in clause-initial position (see example 2a), no matter where the constituent occurs in the corresponding declarative clause. In *wh-in-situ* languages like Mandarin Chinese, by contrast, the question word surfaces in the same position as in a declarative clause, which is the post-verbal position for direct objects (see example 2b):

- (2) a. **What** did Anna buy?
 b. Húfēi mǎi-le **shénme**
 Hufei buy-PRF what
 ‘What did Hufei buy?’ (Bayer & Cheng 2017, p. 2)

Variability also arises within a given language when speakers have more than one linguistic option to express a concept or interpret a linguistic object. Such variability can arise through dialectal or regional preferences, context-dependent registers (though these will not be addressed in this paper), or it can reflect true optionality (not restricted to a certain dialect or context) in a language. Such within-language variability can be found between (sub)groups of speakers – a case of inter-individual variability, which arises because linguistic structures and their associated meanings must be conventionalized across members of a group and adapted to context and communicative goals for successful communication. It can also be found within a speaker, an instance of intra-individual variability.

An example of inter-individual variability is auxiliary selection in Italo-Romance: The perfect auxiliary in many of these varieties can surface either as *have* or *be*. The choice is conditioned by different factors across geographical areas, e.g., by the person features of the grammatical subject or the argument structure of the lexical verb, sometimes in interaction with tense/aspect specifications in the clause (see Amato 2023 for an overview). Intra-individual variability can be observed, e.g., in English in the grammatically optional operations of *Heavy NP-shift*, illustrated in example 3a, where prosodically heavy/long constituent can be displaced to the right edge of the clause, or by *quotative inversion* (Collins 1996, example 3b), where the auxiliary inverts with the subject after a quote.

- (3) a. John gave to Mary [a book about the richest man in the world].
 b. [“Let’s go to the movies”], said John.

2.2. Linguistically Meaningful Variability vs. Noise

Another important distinction is that between *linguistically meaningful variability*, which results either from the workings of the mental grammar itself, or else from language-external, processing-related cognitive or biological factors, as opposed to *random noise*, which is linguistically irrelevant, and which can be explained neither by the grammar, nor by systematic language-external factors.

In cases of *linguistically meaningful variability*, the set of cognitive principles constituting the linguistic competence provides more than one option for realizing the same content in the same context. This type of variability is grammatically relevant in that it allows for valuable insights into the workings of the underlying grammar and the processor, i.e., into the linguistic competence of speakers. It comprises instances of systematic and predictable variability within and between individuals, but, of course, also across languages.

Linguistically meaningful variability can arise directly from internal grammatical mechanisms, without imposing any noticeable additional or reduced demand on the processor.

Such instances typically arise when different components of the grammar (say, syntax and prosody) impose competing constraints on the linguistic output, such that either one or the other, but not both, can be satisfied; cf. the discussion of OT-grammars and tied constraints in section 3.1.3. Here, we discuss but one prominent example, namely the variable word order of focused indirect objects in the German midfield (Büring 2001). Both variants in example 4, which differ in the linear order of the direct object (DO) and the indirect object (IO), are equally licit answers to the question *Who did Gabriel give the book to?*, with no discernible differences in meaning or processing effort.

- (4) a. Gabriel hat [ANNA] [das Buch] gegeben.
 Gabriel has Anna the book given
 ‘Gabriel gave the book to ANNA.’
 b. Gabriel hat [das Buch] [ANNA] gegeben.
 Gabriel has the book Anna given
 ‘Gabriel gave the book to ANNA.’

The optional word order follows from the fact that both linearizations constitute suboptimal attempts at satisfying general constraints on prosodic and syntactic complexity. Example 4a satisfies the syntactic constraint on basic word order IO>DO in the German midfield (Lenerz 1977), but it violates the prosodic constraint that the nuclear pitch accent be realized immediately before the clause-final verb. In contrast, example 4b satisfies the prosodic constraint at the cost of scrambling to a marked syntactic word order DO>IO.

Linguistically meaningful variability can also be systematically driven by language-external factors, such as predictability or processing effects. Predictability-driven effects show up in the form of inverse form-meaning correlations between the presence of additional formal marking and the relative unexpectedness of a form or function. The more unlikely a given form or function, the more likely some extra grammatical marking; see section 3.2 for more discussion; for processing-based word order phenomena see section 3.4. Another example is the optional prefocal placement of *ornamental pitch accents*, which is conditioned by rhythmic and extra-linguistic prosodic factors (Büring 2007). Lastly, Pregla (2023) shows that individuals with acquired language impairments (aphasia) exhibit greater and less consistent variability, both within and between participants, when compared to unimpaired controls. This suggests an impairment at the processing level rather than a deficit in the underlying mental representations. The latter would lead to stable deficits, resulting in consistent impaired behavior of an individual across tasks and over time.

Linguistically meaningful variability must be distinguished from variability that consists of *random noise*, which cannot be explained by systematic language-internal or -external factors. Noise comprises classical performance errors, e.g., speech errors, and it does not provide insights into the underlying mental systems constituting our linguistic competence. A central methodological and empirical tenet for linguistic research is therefore to clearly differentiate between noise in the data and linguistically meaningful variability; cf. section 4.1 for more discussion.

2.3. Hidden Variability

The linguistic study of variability often focuses on easily observable variability in surface strings, the explanation of its existence, and on regularities found in it. A much less discussed instance of variability in natural language is what we refer to as *hidden variability*.

Hidden variability is instantiated by surface-identical linguistic structures with different underlying grammatical representations between languages, or even between and within speakers of the same language. It is often found with frequent construction types or with constructions of high communicative value. For example, there is evidence for the validity of both the raising and the matching analysis for *relative clauses* (RCs), even within the same language (Sichel 2014).¹ Likewise, Barros & Vicente (2011) argue that we need both an ellipsis and a multi-dominance approach to account for the properties of *Right Node Raising* in English, a coordination construction in which a constituent at the right edge of the sentence (bold-faced in example 5a) seems to be shared by (i.e., interpreted in) all conjuncts. While only the ellipsis approach can capture morphological mismatches (e.g., in the form of pronouns), well-known from VP-ellipsis, in the shared material (see example 5b with the elided string in <>), only multi-dominance accounts offer a way to implement cumulative agreement as in example 5c (where the shared finite verb surfaces in its plural form even though the agreement controllers in both conjuncts are singular).

- (5) a. Alice has written, and Beatrix has read, **a book**.
 b. Alice has slept in **her** office, but Bob will not <sleep in **his** office>.
 c. Alice is proud that Beatrix <has>, and Claire is happy that Diana, **have/ *has** travelled to Cameroon. (Barros & Vicente 2011, p. 1-4)

In the domain of word order, it is well-established that *verb-initial order* can be derived by head movement of the verb, i.e., the verb moves alone, or by VP-fronting, i.e., the verb moves as part of a larger syntactic constituent (see Clemens & Polinsky 2017 for an overview). Furthermore, different underlying structures for surface-identical *Serial Verb Constructions* are found both within and across languages (Amaechi et al. 2023).

Whereas instances of *hidden variability* are often found in the domain of syntax, the phenomenon shows up in other domains as well. In morphology, the apparently unified class of preverbal subject markers in Bantu divides into two subkinds, namely agreement markers vs. incorporated pronouns (Bresnan & Mchombo 1987). As an agreement marker, the subject marker repeats the person-number features of the independent subject NP on the verb. As an incorporated pronoun, the subject marker *is* the subject constituent. In phonology, acoustically indistinguishable cross-height vowel pairs (e.g. /o/-/u/) in several Niger-Congo languages are in complementary distribution with respect to vowel harmony and are thus underlyingly featurally distinct (Rose et al. 2023). Although acoustically indistinguishable, they either exhibit ATR-harmony with other +ATR-vowels (/o/) or other -ATR-vowels (/u/). Finally, at the syntax-semantics interface, bare NP arguments may have different underlying representations as *ι*-definite, $\exists$ -quantifier, or predicates of type *et* (Jenks 2018; Driemel 2023; Philipp 2023).

¹On the raising analysis, the head noun *letter* originates inside the RC (underlined in example 1a), from where it raises to its surface position. On the matching analysis, the relative head noun has a matching RC-internal counterpart that undergoes movement to a left-peripheral position inside the RC and remains unpronounced; cf. example 1b. See Sauerland (2003) for discussion of these two types of approaches to RCs.

- (1) a. I read the **letter**_{*i*} [that Anna wrote letter_{*i*}].
 b. I read the letter_{*i*} [~~letter~~_{*j*} that Anna wrote letter_{*j*}].

The existence of *hidden variability* has major consequences for theories of grammar. For one, it explains why seemingly surface-identical constructions in different languages come with different properties. More importantly, it also directly accounts for inter-individual variability in judgments about such construction types, and also why some constructions seem to have mixed properties; see, e.g., den Dikken et al. (2000) on different subtypes of surface-identical pseudoclefts in English. Although *hidden variability* is often viewed as less desirable compared to uniform analyses, ignoring it as an analytical possibility has major drawbacks: First, it can result in the false unification of underlyingly distinct phenomena in formal modeling. Second, it may contribute to premature conclusions about typological patterns based on the non-exhaustive study of superficially similar structures (see Sande et al. 2019 on SAuxOV order in West African languages, which can result from mixed headedness or from head-initiality + object shift). Third, it can have the effect that the emergence of certain patterns is attributed to universal linguistic factors, rather than to language-external causes of language change (e.g., processing ease, language contact). In sum, the controlled study of *hidden variability* may provide an explanation for the existence of conflicting diagnostics for surface-parallel forms.

3. ACCOUNTING FOR VARIABILITY IN GRAMMATICAL THEORIES

A central task of any grammatical theory is to account for what is a possible human language (Chomsky 1965; Newmeyer 2005). Many grammatical theories also aim at a principled distinction between probable, i.e., typologically common languages or constructions, on the one hand, and typologically marked languages or constructions, on the other, thereby identifying some cross-linguistic limits of variability. Importantly, grammatical theories also need to be flexible enough to account for the observable linguistic variability. Different grammar-theoretical approaches have taken different perspectives on variability and its limits in natural language. The existing treatments mainly differ in the importance assigned to language-internal variability, the importance assigned to language-external factors in the shaping of grammars (i.e., formal vs. functional analyses), and in whether they take a categorical or probabilistic approach to language. Below, we will briefly summarize some prominent grammatical theories and highlight which types of variability they capture.

3.1. Generative Approaches

Generative theories of grammar aim at identifying sets of grammatical rules and principles that allow for the derivation of all the grammatical structures of a language, and only these. This section investigates both deterministic (both derivational and representational) and constraint-based *optimality-theoretic* grammar formalisms for their ability to account for intra-, inter-, and cross-linguistic variability.

3.1.1. Derivational Models. The main goal of formal grammar theories since Chomsky (1965) is to provide a model of the human language faculty by defining an internalized mental system of grammatical rules, principles, or constraints, which together constitute the mental *I(nternal)-grammar*, or a speaker's *linguistic competence*. Derivational models generate the grammatical structures of languages by applying ordered sequences of rules to some underlying input structure. Such systems should successfully distinguish grammatical from ungrammatical structures in any given language, and they should account for the ob-

servable cross-linguistic variability. This variability is limited by (possibly universal) general constraints that define a class of linguistic structures as (un)available in any language.

As mentioned in section 1, Chomsky (1965) abstracts away from surface noise and inter- and intra-individual variability to characterize the underlying linguistic competence of speakers by focusing on the invariable primitives. Chomsky’s (1965) distinction between *D(eep)-* and *S(urface)-Structure* allows for a large degree of inter- and intra-speaker consistency and cross-linguistic uniformity at the thematic structure-building layer of D-structure, in spite of variability on the surface. This underlying uniformity can be masked by subsequent language-specific rules or by grammar-external performance-based factors, such as register/style, constraints on the articulatory system, processing requirements, or socio-linguistic factors. Examples of such intra- or inter-individual variability in English are the grammatically optional operations of *Heavy NP-shift* and *quotative inversion* (Collins 1996), see example 3. D-structure was also considered the locus of the Greenbergian universals (Greenberg 1963), with apparent violations attributed to language-specific rules (Newmeyer 2005). An example of a postulated universal uniformity is Kayne’s (1994) claim that all languages come with SVO-order underlyingly, no matter their basic surface order.

Subsequent generative work modeled cross-linguistic variability within the *Principle and Parameters (P&P)*-framework (Chomsky 1981, 1986), which derived structural differences between languages from the interaction of innate universal principles with language-specific parameter settings. The universal principles are taken to be encoded in a highly structured, rich mental grammar – called “Universal grammar” (UG) – common to all languages, thereby accounting for cross-linguistic similarities.² Some parts of UG are, however, underspecified in that several options are made available; these options are called parameters. A major part of language acquisition consists in setting these parameters, i.e., in choosing the option that is present in the learner’s input in their language community. For example, the *pro-drop parameter* (Perlmutter 1971; Rizzi 1982) was supposed to determine whether pronominal subjects are overtly realized (English *I phoned*) or not (Italian *Ho telefonato*). Relevant works in this area include Manzini & Roussou (2000) on the *pro-drop parameter* and its implications for cross-linguistic variability; Baker’s (2001) parameter hierarchy to account for typological (un)markedness; Bobaljik (2002) and Baker (2008) on the workings of parameters at the syntax-morphology interface, including variability in agreement systems; Roberts & Holmberg (2010) on a *Minimalist* extension of *P&P*-theory; and Newmeyer (2005, ch. 2.3) for a critical evaluation regarding its coverage of cross-linguistic variability.

It quickly emerged that the macro-parameters from *P&P*-theory were too coarse-grained to account for the observable differences even between typologically closely related languages like English and French. Subsequent work thus focused on the identification of micro-parameters through the study of dialects, which only differ in few structural features (cf. Kayne’s 1996 *micro-comparative* approach). Some representative examples are the *Syntactic Atlas of the Dutch Dialects* (SAND, Barbiers et al. 2007), the *Syntax of Hessian dialects* (SyHD, Fleischer et al. 2017), and Poletto (2000) on Northern Italian dialects.

The *Minimalist Program* (Chomsky 1995) abandoned macro- and micro-parameters in favor of lexical specifications. Syntactic differences between languages are now attributed to different feature specifications of functional heads (i.e., minimal structural units representing non-lexical categories), cf. the *Borer-Chomsky Conjecture*, (Baker 2008), or to

²The universal principles include, e.g., the *Projection Principle*, *Subjacency*, the *Empty Category Principle*, and the *Binding Principles* (see Haegeman 1994 for definitions and relevant phenomena).

language-specific spell-out rules for abstract syntactic categories at the interface with phonetic articulation. For instance, the difference between *wh-ex situ* and *wh-in situ* languages (see example 2 in section 2.1) follows from whether spell-out realizes the higher or the lower copy of a movement chain (Groat & O’Neil 1996). In addition, cross-linguistic differences may follow from language-specific post-syntactic rules such as fusion or merger (Halle & Marantz 1993). The only remaining (innate) universal feature of human grammars is the applicability of the structure-building mechanism *Merge* (Hauser et al. 2002).

Both the *P&P*-framework and the *Minimalist Program* can easily deal with inter-individual variability. As the parameters or functional feature specifications are set individually in the process of language acquisition, there could in principle be as many combinations as there are human speakers or feature combinations of functional heads. Newmeyer (2005, p. 83) points to Kayne (2000, p. 8), who calculates that only thirty-three binary-valued parameters are required for deriving “some number substantially greater than five billion” grammars, i.e., close to one grammar for every two human speakers on earth. Of course, language input in language acquisition and close speaker contact in communication will still ensure a rather homogeneous feature specification in individual language communities.

By contrast, intra-individual variability has only played a minor role in classic generative analyses of natural language grammar. Once an *I-grammar* is acquired, speakers are (tacitly) expected to show a consistent linguistic behavior, with only little surface variability. Apparent violations of grammatical constraints have often been attributed to random performance errors, which in turn are attributed to language-external factors, such as tiredness, distraction, or other cognitive or processing-related facts (Chomsky 1965). However see, e.g., Adger (2016) on how to integrate intra-individual variation (in morphosyntax) into *Minimalism*, which also predicts the frequencies of the variants in actual language use.

3.1.2. Representational Models. Representational generative models such as *Lexical Functional Grammar* (see Dalrymple 2001; Bresnan et al. 2016 for overviews) or *Head-Driven Phrase Structure Grammar* (Pollard & Sag 1994; Müller 2007) only assume one structural layer of representation, where grammatical relations between different parts of the clause are not modulated by movement, but via feature specifications and feature unifications. Being so-called “lexicalist” theories, HPSG and LFG postulate information-dense lexical entries; in HPSG, these entries encode phonological, morphosyntactic, and semantic properties, including, e.g., the combinatorial potential of each lexical item. This allows the models to represent variability across and within languages via lexical entries. Cross-linguistic and inter-individual differences are modeled by different feature specifications on terminal elements (see, e.g., Sio & Song 2015 for an HPSG-account of variability in the coding of definiteness in Mandarin vs. Cantonese). It is less clear, though, how such models would deal with intra-individual variability, as they are quite deterministic in distinguishing between grammatical and ungrammatical structures on the basis of different feature specifications; but see Machicao y Priemer et al. (2023) for an integration of register variability in HPSG.

3.1.3. Optimality-Theoretic Approaches. In contrast to derivational and representational models, competition-based approaches using violable constraints such as *Optimality Theory* (OT, developed out of harmonic grammar approaches, see, e.g., Legendre et al. 1990) are inherently better suited to account for optionality in grammar and for cross-linguistic and inter-/intra-individual variability in linguistic behavior. This is because they do not conceive of grammar as a deterministic system of absolute rules, but as an evaluation matrix of a set

of universal but violable ranked constraints. These must be satisfied to the best possible degree. Thus, a given linguistic form may still be the best option available even if it does not satisfy all constraints on well-formedness (McCarthy 2002; Prince & Smolensky 2004).

In OT-systems, variability is accounted for in terms of reranking of universal constraints, which gives rise to a factorial typology (i.e., every ranking of the constraints should represent a possible natural language). This property of OT is usually discussed and illustrated for cross-linguistic variability. For instance, the difference between *wh-in situ* and *wh-ex situ* languages (see example 2) can be captured by different relative rankings of two types of constraints: A markedness constraint that requires *wh*-elements to occupy a left-peripheral position, and an economy constraint that prohibits displacement of the *wh*-element from its normal position. Languages where the economy constraint outweighs the markedness constraint exhibit *wh-in situ*, whereas the opposite constraint ranking yields *wh-ex situ*, see Müller (2000, ch. 2.1) and references cited there for proposals along these lines. Constraint re-ranking has also been applied to account for regional/dialectal microvariation (see, e.g., Legendre 2010 on person-based auxiliary selection in Italo-Romance, e.g., Ariellese *Ji so' fatte na torte* 'I **be** made a cake' vs *Esse a fatte na torte* 'She **have** made a cake'), see section 2.1, and for inter-/ intra-individual variability (see Choi 1999; Büring 2001). Büring (2001) presents an analysis of language-internal variability in the realization of indirect object focus in German, which necessarily violates either a syntactic or a prosodic constraint, cf. example 4. The integration of within-language variability in OT is facilitated by various technical means that can derive optionality (see Müller 2003), e.g., constraint ties by which two or more constraints are not ranked with respect to each other, and are thus equally important. Alternatively, inter- or intra-individual variability has been accounted for in stochastic OT-analyses (e.g., Boersma & Hayes 2001; Keller & Asudeh 2002), which also represent the frequencies with which various structures are used. Variability is derived from the fact that the relative ranking of some constraints is probabilistically underdetermined: there is a (slight) chance for either ranking. The evaluation component will then give out one form as optimal on one ranking, and another form on another ranking; see Salzmann & Seiler (2010) for a stochastic OT-analysis of microvariation in Alemannic relativization.

3.2. Probabilistic Approaches

The introduction of tied and probabilistically weighted constraints in OT-grammars is a marked departure from the classical deterministic approach of generative grammar. The idea that a given linguistic construction type is more or less likely to occur cross-linguistically or between individual speakers of a language is also found in other approaches to the structure of language. *Frequency-* or *predictability-*based approaches to language are found in Hume (2008); Haspelmath & Karjus (2017); Haspelmath (2021), i.a. In considering the relation of form and meaning in general functional-typological terms (cf. section 3.4), these approaches assume a trade-off between explicitness and parsimony in the signal in order to account for optimal coding patterns. In a nutshell, more predictable meanings correlate with less or zero-marked linguistic structures, whereas less predictable meanings correlate with more variable and marked structures; cf. Kurumada & Grimm (2019) for an overview. Such form-meaning correlations can show up within a given language, as has been found, e.g., in connection with language-production studies; see, e.g., Rose (2017) on variability in the pronunciation of the plural suffix *-s* in New Zealand English, and Bürki et al. (2010) on variability in *schwa deletion* in French. In addition, they also emerge at the cross-linguistic

level, where they give rise to generalizations of typological markedness (see Haspelmath 2021 for a recent overview). Finally, predictability and typological unmarkedness also have an effect on language-internal coding preferences whenever a given marking is grammatically optional. See, e.g., Kurumada & Jaeger (2015) on optional case marking in Japanese, and Haspelmath & Karjus (2017) on grammatically optional PL(ural) marking in Welsh. The latter show that the presence or absence of explicit PL-marking is inversely correlated with the cross-linguistic frequency of the plural concepts expressed, such as stereotypically plural [peas] (no PL-marking in Welsh) vs. stereotypically singular [father] (PL-marking). See Kurumada & Grimm (2019) for a replication in an artificial language learning paradigm.

Following pioneering work on bidirectional OT in form-meaning pairing in production and comprehension, where the optimization procedure simultaneously considers both listeners' and speakers' perspectives (Blutner 2000, Benz & Mattausch 2011), Bayesian probabilistic models are increasingly employed by semanticists in exploring variability in the form-meaning interface; cf. Erk (2022) and Bernardy et al. (2022) for a recent overview and a textbook introduction, respectively. A popular representative of this line of research is the *Rational Speech Act (RSA)*-model (e.g., Frank & Goodman 2012; Goodman & Lassiter 2015, and in particular Scontras et al. 2025 for an online manual). RSA is a computational model that utilizes a recursive reasoning procedure to assess the pragmatic listener's probability that a speaker meant to convey a given meaning *M* by uttering a linguistic form *L*. The model incorporates prior probabilities (*How likely is M to hold true in the world?*), economy considerations (*How complex is L?*), and general Gricean (1975) reasoning over the speaker's rational and efficient communicative behavior to predict the probability of *M*. It has been successfully used to model the interpretation of gradable adjectives (Lassiter & Goodman 2017) and vague numerical approximators like *around* and *between* (Egré et al. 2023); also see Schreiber & Onea (2021) for critical discussion of the limits of RSA-models. Empirical support for Bayesian reasoning in interpretation comes from experimental semantic research that shows that the (un)availability of inverse quantifier scope readings within and across languages differs gradually, not categorically; see Philipp & Zimmermann (2023) on the graded availability of relative quantifier scope in English and German.

3.3. Competing Grammars

Another model of grammar that can directly account for inter-individual and cross-linguistic variability in linguistic behavior is the *Competing Grammars* model (cf. Kroch 1989; Yang 2000; see also section 4.2). This model assumes that the language-acquiring infant will entertain input-based hypotheses on what the grammar of its input language will look like. Specifically, the model assumes that the infant will select that grammar from the finite set of possible grammars that is most compatible with the data, and hence most probably the correct grammar. However, the model also allows for the possibility that speakers acquire two grammars *G1* and *G2* with equal probabilistic weighing, so that it is not clear which grammar is correct. In such cases, speakers would have two structural options at their disposal for expressing a given meaning: *G1* outputs form1, whereas *G2* outputs form2. However, such situations tend to be diachronically unstable. They typically result in specialization, where the two forms acquire different meanings, or in the loss of one structural alternative, cf. Geeraerts (1999). In place of outputting different surface forms, the two grammars may also assign different structural analyses to the same input string, thereby instantiating a case of *hidden variability*; see section 2.3.

3.4. Functionalist Approaches

Functionalist approaches take the performance-related aspects of language as their starting point. The central claim is that the structure of natural languages is determined by the need to convey information in an optimal way, where optimality is mainly determined by processing considerations; see, e.g., Hawkins (1999). From this perspective, grammar can be considered “a conventionalization of performance preferences” (Newmeyer 2005, p. 119), as captured in the *Performance-Grammar Correspondence Hypothesis* (Hawkins 2004).

This approach to grammar can easily capture language-internal variability. It predicts variability to be possible whenever two competing structures put roughly the same burden on the processor, such as in *Jeanne talked [to Carl] [about Anne]* vs. *Jeanne talked [about Anne] [to Carl]* (Newmeyer 2005, p. 121), or when the two structures violate opposing constraints that cannot be simultaneously satisfied (Hawkins 2004): The lack of relative clause (RC) extraposition in English in example 6a, from Francis (2010, p. 36), satisfies locality (small distance between head noun and RC), whereas RC extraposition in example 6b satisfies prosodic constraints (heavier constituents follow lighter ones, Wasow 1997).

- (6) a. New sets that were able to receive all the TV channels soon appeared.
- b. New sets soon appeared that were able to receive all the TV channels.

Hawkins (2004) proposes three general processing constraints on linguistic structures. First, *Minimize Domains* is a locality constraint that captures a preference for structural proximity of syntactically and semantically dependent elements. This constraint becomes manifest, e.g., in the privileged status of grammatical subjects in syntactic processes (*wh*-extraction, relativization, ...) in Keenan & Comrie’s (1977) *Noun phrase accessibility hierarchy*: Grammatical subjects are structurally prominent and hence the closest available arguments for the relevant structural operations. *Minimize Forms* is an economy constraint and refers to a preference for minimally complex forms to facilitate production and parsing effort (parsimony). Finally, *Maximize Online Processing* refers to a preference for maximizing the set of properties that are assignable to each item X as X is processed. This constraint counteracts *Minimize Forms* and strives for maximum explicitness in the signal. Functionalist analyses can then model language-internal variability between or within individuals as a trade-off between different processing-related constraints, where some of these constraints may be correlated with differences in general cognitive endowment (e.g., working memory, attention span, processing speed, etc.), often as a function of age or neurological disposition.

Functionalist approaches to grammar are well suited to account for typological (un)markedness (Newmeyer 2005). To repeat, a central assumption of functionalist approaches is that “[g]rammatical conventions are ‘frozen’ processing preferences” (Hawkins 1999, p. 279). One such preference is the preference for consistency in linearization, as manifest in the consistent head-first or head-final patterns in head-complement orders: Consistent head-first languages like English have a general preference for lighter before heavier constituents, whereas head-final Japanese exhibits the opposite preference. Another example of cross-linguistic unmarkedness is the above-mentioned preference for subjects over other nominal arguments in syntactic processes (Hawkins 2004).

The assumption that grammatical structures are essentially processing-driven raises a final question regarding cross-linguistic variability. How to account for the observable cross-linguistic variability in language patterns if the cognitive mechanisms involved in language processing are universal? Haspelmath (2021) attributes this to a trade-off between the competing pressures of parsimony and explicitness: “While some languages in some patterns

show fully explicit coding, and some languages in some patterns show no overt coding (i.e. maximal parsimony), many languages make an efficient compromise, where only the less predictable information gets overt coding, while the more predictable information is left uncoded [...] Counter-efficient patterns are virtually unattested.” (ibid., p. 629).

3.5. Construction Grammar

Construction Grammar (CxG) is a largely functionalist approach to grammar (cf. Goldberg 1995; Hilpert 2014; Fried & Nikiforidou 2025 for an introduction). Constructions are lexically stored items that pair syntactic structure with semantic and phonological information; these items can be word-sized but also multi-word expressions or even phrasal chunks. Constructions are connected through an inheritance hierarchy in the lexicon, which is thus a highly structured network. The mental construction of this network (viz., the establishment of vertical and horizontal links, and their strength) as well as the choice of a particular construction in a given context is also influenced by several non-grammatical factors: (a) the frequency of a construction in the communicative environment of children and adult speakers: The more frequent a structure, the more conventionalized and thus stable it is, while less frequent structures are more prone to give rise to variability and to undergo (diachronic) change (e.g., Ellis & Ferreira-Junior 2009; Bybee 2010); (b) the processing ease of a constructions, with a preference for constructions that are easier to process; (c) productivity (Goldberg 1995); and (d) social factors. This makes CxG a usage-based model of language and also allows for the integration and explanation of variability on all levels (see Coleman & Noël 2025 for discussion): Which constructions are stored in the learning process and which links between them are established highly depends on the input (e.g., the frequency of occurrence with(out) different lexicalisations) and thus differs between languages and speakers. For example, Croft (2001) adopts an explicit typological perspective in his CxG account of universal cross-linguistic tendencies and language-specific variability. Boas (2003) presents a case study of resultative constructions in English and German and shows how variability emerges in two subtypes of resultatives. Leino & Östman (2005) present a CxG-based analysis of within-language variability in Finnish (e.g., case marking of arguments of perception verbs, formal vs. informal person marking on verbs) including cases of free variation and intra-individual variability. However, see Adger & Trousdale (2007) on how certain principles of CxG (e.g., Goldberg’s 1995 ‘No synonymy’) may constitute a challenge for the integration of intra-individual variability in CxG. Variability and ambiguity in input during language acquisition have also been used to model language change in CxG, see, e.g., Traugott & Trousdale (2013).

3.6. Summary

The major grammatical theories can all accommodate and account for observable inter-individual, intra-individual, and cross-linguistic variability in linguistic behavior, if not all to the same extent. Formal approaches model variability in terms of different feature specifications or differences in probability distributions, while functional approaches emphasize competing pressures in processing or differences in the input frequency of a construction. In the end, any successful theory of grammar must integrate elements of both formal and functional approaches; cf. Thomas (2020).

4. Effects of Intra- and Inter-Individual Variability

This final section discusses some other effects of intra- and inter-individual variability on language, and its consequences for grammatical modelling. Section 4.1 investigates the importance of variability in quantitative data for theory-building. Section 4.2 highlights the importance of variability for language acquisition and language change.

4.1. Variability in Quantitative Data

Quantitative linguistic experiments typically exhibit a large amount of inter- or intra-individual variability in the data. This variability was standardly regarded as spurious noise, cf. section 2.2, to be dealt with by averaging over participants in the statistical analysis. However, closer statistical scrutiny (Winter & Grice 2021; Vasishth 2023; Vasishth et al. to appear) often brings to light interesting consistencies in inter- and intra-individual variability, typically in the form of bi- or multimodal distributions. These provide an important window into the functioning of the linguistic system, as they are often indicative of systematic language-internal variability in grammar or interpretation (participants chose one of two grammatical or interpretive options). Alternatively, they point to the existence of different speaker subgroups (inter-individual variability). Members of different subgroups may be endowed with minimally different grammars, or subgroup formation is conditioned by language-external factors, such as age or general cognitive abilities.

Bi- or multi-modal distributions have been observed in both offline and online experiments: De Vaugh-Geiss et al. (2018) found two speaker groups in an offline experiment on the interpretation of focus-background cleft sentences in German. One group interpreted clefts as exhaustive, whereas the other did not. The optional flouting of the exhaustivity inference supports a pragmatic analysis of cleft exhaustivity. Zimmermann et al. (2020) replicated these bimodal findings for focus clefts in English, French, Russian, and Hungarian; and De Vaugh-Geiss (2021), for focus fronting in typologically unrelated Akan (Kwa). Next to pointing to a possible cross-linguistic universal, these findings are important because they may provide an explanation for why it has been difficult to assess the status of focus clefts as either [+] or [-] exhaustive in formal semantic literature. To give another example, Brasoveanu & Dotlacil (2015) identify three groups of speakers in a forced-choice experiment on quantifier scope in English (surface bias, inverse bias, no bias) and interpret their findings in terms of dialectal variation. Alternatively, the data could be indicative of inter-individual preferences for the pragmatic resolution of a generally available scope ambiguity; see, e.g., Philipp & Zimmermann (2023) on quantifier scope resolution in English and German, and Gil (1982) on inter-individual variability in scope-taking with numerals.

In experimental work on syntactic variability, Fanselow et al. (2024) find a trimodal partitioning in the data. In an acceptability study on the availability of (partial) *wh-copying* in German and Dutch, they investigate structures with simple, with right-complex, and with left-complex copying, here illustrated for German after Fanselow et al. (2024):

- (7) Wen (von den Studenten) denkst du wen (von den Studenten) sie mag?
who (of the students) think you who (of the students) she likes
‘Who (of the students) do you think that she likes?’
- (8) a. wen ... wen [simple]
b. wen ... [wen von den Studenten] [right-complex]
c. [wen von den Studenten] ... wen [left-complex]

Using a Gaussian finite mixture model, Fanselow et al. (2024) found three types of grammars (*wh*-doubling and right-/left-complexity, *wh*-doubling and right-complexity, no *wh*-doubling), but there was no evidence for grammars with *wh*-doubling and a preference for left-complexity (Fanselow et al. 2024). The results are of theoretical interest as they identify clear limits of variability and provide potential evidence for the copy-theory of movement (Chomsky 1995), which involves the copying of a *wh*-phrase, placing it in a higher position, and deleting the original item. The copy-theory predicts only the pattern with doubling and preference for left-complexity to be absent, “as it is possible to partially copy a *wh*-constituent but not to add material to the original” (Fanselow et al. 2024, p. 42).

The identification of systematic inter-individual variability and its sources is also of importance in online studies. For instance, recent analyses of brain wave data (EEG) highlight systematic inter-individual differences that, if ignored and averaged across participants, could result in null findings. Kim et al. (2018) tested for a trade-off between the N400 and P600 components in sentence comprehension. They found that participant subgroups either displayed a more prominent N400- or P600-response to the same stimuli, where the trade-off was related to individual working memory capacities. The authors conclude that it is the capacity of the working memory that is limiting whether sentence comprehension relies mainly on accessing or integrating a word’s semantic representation, indexed by the N400, or on structural processing, indexed by the P600. Rabovsky et al. (2024) identified similar individual differences in the prediction of sentence meaning and the prediction of the next word (surprisal) by using computational modeling and EEG data. In the majority of participants, the N400 response was generated by semantic updating processes or by a combination of semantic updating and surprisal, while in around one fifth of participants the N400 was mainly driven by surprisal, measured as the prediction error of the next presented word. Averaging across these three participant types would once again obscure the underlying processes.

Individuals differ not only on the processing side but also in how they produce speech. Bürki (2018) gives an overview of how variability at the word level provides crucial information on the cognitive architecture of the language production system. To determine reliable inter-individual differences, it is important that the individual speakers’ productions are stable. Recently, individual differences in picture naming speed have been shown to be reliable between sessions and tasks (Fuhrmeister et al. 2024). In prosody, when producing major prosodic boundaries, some speakers consistently use a combination of all typical prosodic cues (for German: higher pitch and longer duration of the final syllable before the boundary, followed by a silent pause), while others use only two of them (Huttenlauch et al. 2021). Although variability within participants is limited, each individual production will still differ from the other. This follows because—despite phonology being an abstract system of discrete computations—the motor speech organs generate speech output in a continuous manner, which is subsequently interpreted by the listener as having discrete meaning. For a discussion of how nonlinear dynamic models can relate discrete phonology and continuous phonetics, see, e.g., Gafos & Benus (2006) and Roessig et al. (2019).

4.2. Variability in Language Change and Acquisition

A major driving force in language change is language acquisition. Dating back to Lightfoot (1979), the idea is that infants faced with ambiguous or opaque speech input perform a reanalysis (or neoanalysis, Traugott & Trousdale 2013). In the long run, such reanalyses

can induce language change. Building on this, Weiß (2024) summarizes the important role of variability as both an important outcome and a trigger of language change: Synchronic patterns of inter-individual variability in pronoun ordering in German can be explained as remnants of different stages in the so-called *Pronominal Cycle* (Weiß 2015). Specifically, there are three groups of speakers in contemporary German that exhibit different pronoun orderings without any sociolinguistic implications. Eventually, some of these variants may spread to other groups or communities, or change over time, where such changes may be caused by sociolinguistic factors and/or processing factors. Weiß (2024) further argues that variability can also function as the trigger of language change, namely in the form of *hidden variability* discussed in section 2.3. An underlying structural ambiguity of the same surface string can trigger structural reanalysis, and hence language change from one generation to the next. Typically, this reanalysis takes place when the child acquiring their system of mental linguistic rules is confronted with structurally-ambiguous surface strings, e.g. example 9 concerning the grammaticalization of the German preposition *seit* ‘since’ in empty-headed RCs into an embedding complementizer (Weiß 2024, p. 33f.).

- (9) a. **sît** **des**, **daz** ich von lande schiet [Middle High German]
 since that.GEN that I from country departed
 ‘since I departed from the country’
 b. $[_{PP} \text{ sît } [_{NP} (\text{des/dem}) [_{CP} (\text{daz}) \dots]]] \rightarrow [_{CP} [\text{C } \textbf{sît}] \dots]$

A fundamental question for the theory of syntactic change in frameworks involving parameter setting is how the parameters are set in language acquisition. Biberauer & Roberts (2017) argue that the input cannot be *reanalyzed* from the infant learner’s perspective, and the parameters cannot be *reset*. Rather, they must be *set* initially. Diachronic language change is therefore based on parameter *optimization*. In addition to a possible innate endowment of infants and their linguistic experience (primary linguistic data), non-domain-specific cognitive optimization principles are also at work. The parameter values thus emerge from the interaction of these three factors, which can eventually lead to a syntactic change. In the framework of *Competition Grammar* from section 3.3, language change is modeled as a change in grammar distributions over time (Yang 2018, 2000). Based on strictly biological general learning theories, language acquisition is viewed as a “variational process in which the distribution of grammars changes as an adaptive response to the linguistic evidence in the environment” (Yang 2000, p. 234). Ultimately, this variational model allows for individual differences, as the statistical makeup of the input can differ for each individual learner, opening up the possibility that each learner may acquire their target grammar on their own unique timeline. The model assumptions have been validated with historical data, such as the loss of the verb-second phenomenon in Old French and Old English (Yang 2000).

All of these theories have in common that they treat first language acquisition and language learning (through contact, migration, sociolinguistic pressure, etc.) as critical for language change; for change in heritage languages, see D’Alessandro et al. (2025). They further hypothesize that processing constraints play an important role as well. Here, biological and other non-linguistic evolutionary forces can lead to the survival of linguistic structures that are less costly than more complex structures. However, in the majority of cases we lack systematic diachronic and synchronic data to validate existing theories. An elegant way of addressing this problem is to test concrete hypotheses on the driving sources of diachronic change (e.g., processing factors) attested in diachronic corpora with actual speakers in synchronic psycholinguistic experiments. De Cesare et al. (2024) investigate

diachronic change in the word order of infinitival complements of German control verbs (e.g., *versuchen* 'to try') from intraposed position (pre-verbal, example 'Julia sagt, dass Fred [den Kuchen zu schneiden] versucht', 'Julia says that Fred the cake to cut tries') to extraposed position (clause-final, 'Julia sagt, dass Fred versucht [den Kuchen zu schneiden]', 'Julia says that Fred tries the cake to cut'; examples from Bosch et al. 2022). They show that intraposition incurs a high processing cost because it involves center embedding and triggers a temporary ambiguity in online processing. This correlates with the fact that, over time, intrapositions have become ever less frequent – so much that actual speakers avoid producing such structures altogether. These findings underscore that processing factors, in addition to sociolinguistic factors or stylistic norms, can affect the variability of linguistic forms, and thereby shape the present-day language use. The integration of corpus-based and experimental methods holds promise for increasing our understanding of the constraints that restrict variability in language use and language change (De Cesare et al. 2024).

Experimental evidence from studies using laboratory artificial grammar learning paradigms supports the assumption that language acquisition can drive language change. In these studies, learners are exposed to language input with variability in syntactic properties, such as the use of determiners or word order. The learning outcomes show interesting differences between children and adults: Adults mirror the distributional patterns of the variable properties, whereas children showed a stronger bias to regularize the most frequent pattern (Hudson Kam & Newport 2005; Culbertson & Newport 2015). It is unclear to what extent these tendencies for regularization in children are affected by cognitive limitations or language universals (Hudson Kam & Newport 2009; Austin et al. 2021), but they may affect other areas, such as language change, creolization, or deaf children's creation of a more consistent sign language based on inconsistent input (Singleton & Newport 2004).

Finally, we would like to stress the importance of input variability for the ontogenetic development of language for individual learners. Somewhat surprisingly, recent experiments on word learning and phonological development have shown the positive effects of exposure to variable productions by multiple speakers. This variability can improve early word learning. Studies using *Habituation-Switch paradigm*, previously found a learning effect in 14-month-olds only when the novel words spoken by a single speaker were phonologically distinct (e.g., /lif/ vs. /neem/, Stager & Werker 1997). However, phonologically similar minimal pairs like /buk/ vs. /puk/ were only learned when spoken by multiple speakers and, therefore, phonetically variable (Rost & McMurray 2009; Höhle et al. 2020). This suggests that variability in speech input facilitates the detection of its invariant properties, which in this case consists of a stable relation between single cues that signal crucial phonetic contrasts. In sum, what may seem like mere noise at the level of individual acoustic parameters is, in fact, beneficial for language acquisition rather than hindering it.

SUMMARY POINTS

1. Variability is an integral part of natural language and needs to be taken into account in models of linguistic competence.
2. Hidden variability can explain conflicting diagnostics for surface-parallel forms.
3. Statistical analyses can reveal consistencies in inter-/intra-individual variability.
4. Variability can be a driving force of language change.
5. Input variability can improve early word learning.

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